Rafael F Barduzzi

PhD candidate

rfbarduzzi@gmail.com

Taxonomy and Evolution of Plants lab
Department of Botany
Washington Luís Highway, Km 235
São Carlos, São Paulo state - Brazil

EDUCATION

2024-present **Ph.D. in Comparative Biology**, Universidade de São Paulo
Title: Unravelling the tempo and mode of independent radiations in the megadiverse genus *Mimosa* L. (Fabaceae, Mimoseae)
Dissertation advisor: Dr. Leonardo M. Borges
Research supervisor (Abroad): Dr. Daniel S. Moen

2018-2023 **B.Sc. in Biology**, Universidade Federal de São Carlos
Undergraduate research: Pollen disparity in the Piptadenia group (Leguminosae): differences between lineages and ecological groups
Advisor: Dr. Leonardo M. Borges

APPOINTMENTS

2025 **Visiting Scholar**, Mississippi State University

2025 **Research Scholar**, University of California Riverside

2024-present **Graduate Researcher**, Universidade de São Paulo

2021-2023 **Undergraduate Researcher**, Universidade Federal de São Carlos

SCHOLARSHIPS

2025 São Paulo Research Foundation (FAPESP) Research Internship Abroad Scholarship

2024-present FAPESP Doctoral degree (Direct) Scholarship

2021-2023 FAPESP Undergraduate Research Scholarship

PUBLICATIONS

Duarte, A. F. T., **Barduzzi, R. F.**, Casas-Restrepo, L. C., Ribeiro, P. G., Queiroz, L. P. D., & Santos, F. D. A. R. D. (2026). Morphological characterization of polyads from *Marlimorimia*,

Pityrocarpa, and *Parapiptadenia* supports the recent circumscription of the Stryphnodendron clade (Caesalpinoideae: Mimoseae). *Acta Botanica Brasilica*, 40, e20260041.

Barros-Souza, Y., Maianne, M., **Barduzzi, R. F.**, & Borges, L. M. (2026). Beyond the Highlands: Climate Drives Evolutionary Connections Between Ancient Neotropical Mountains and Lowland Biomes. *Global Ecology and Biogeography*, 35(2), e70214. https://doi.org/10.1111/geb.70214

Barduzzi, R. F., Liau-Kang, S., Duarte, A. F. T., Santos, F. D. A. R. D., & Borges, L. M. (2026). Morphological innovation and lineage-specific history drive disparification in the aggregated pollen of mimosoid plants. *Annals of Botany*, 137(1), 159-170 https://doi.org/10.1093/aob/mcaf213

Barduzzi, R. F., Barros-Souza, Y., Liau-Kang, S., & Borges, L. M. (2024). Pollen evolution in the Stryphnodendron clade (Leguminosae, Mimoseae). *Brazilian Journal of Botany*, 48(1), 15. https://doi.org/10.1007/s40415-024-01059-y (Part of special issue, *Advances in Legume Systematics 15*)

Liau-Kang, S., **Barduzzi, R. F.**, Lozano Calderón, G., Marega Junior, E., Simon, M. F., Santos, F. D. A. R. D., & Borges, L. M. (2024). More of almost the same: novel pollen descriptions uncover subtle morphological variation in Mimosa L. (Leguminosae, Mimoseae). *Brazilian Journal of Botany*, 47(4), 1071-1090. https://doi.org/10.1007/s40415-024-01033-8 (Part of special issue, *Advances in Legume Systematics 15*)

MANUSCRIPTS SUBMITTED

Barduzzi, R. F., Barros-Souza, Y., Moen, D. S. & Borges, L. M. *Submitted*. Beyond tortoises and hares: Niche lability and a continuum of diversification dynamics underlie the megadiversity of a legume genus.

Barduzzi, R. F., Santos, M., & Borges, L. M. *In press*. Tree thinking and alternative knowledge among undergraduate Biological Sciences students.

PRESENTATIONS

Talks

2026 **Barduzzi, R. F.**; Moen, D. S.; Barros-Souza, Y.; Borges, L. M. A radiation space for *Mimosa*: Evolutionary radiations underlie diversity of one of Brazil's most species-rich plant genera. 2nd Brazilian Congress of Evolutionary Biology, Belo Horizonte (MG), Brazil

2026 **Barduzzi, R. F.** Toward a radiation space of the sensitive plants. V Symposium on Comparative Biology, Ribeirão Preto (SP), Brazil

2025 **Barduzzi, R. F.** Evolutionary radiations in the genus *Mimosa* L. (*Leguminosae*). UCR Center for Conservation Biology's 7th annual EcoSlam, Riverside (CA), USA

2024 **Barduzzi, R. F.** How to identify evolutionary radiations. 11th Meeting on Comparative Biology, Ribeirão Preto (SP), Brazil

2023 **Barduzzi, R. F**.; Liau-Kang, S.; Santos, F. A. R. and Borges, L. M. Pollen disparity in the Mimosa and Stryphnodendron clades (Caesalpinioideae, Mimosoid clade): comparing lineages and ecological groups. 8th International Legume Conference, Pirenópolis (GO), Brazil

2022 **Barduzzi, R. F.** and Borges, L. M. Pollen disparity in Piptadenia group (Leguminosae), 2022, São Carlos. 28th Universidade Federal de São Carlos Congress on Undergrad Research, São Carlos (SP), Brazil

Posters

2025 **Barduzzi, R. F**.; Moen, D. S.; Barros-Souza, Y.; Maianne, M.; Borges, L. The role of leaf morphology, elevation, soil and climate in *Mimosa* L. (Leguminosae) diversification. Botany 2025, Palm Springs (CA), USA

2024 **Barduzzi, R. F**.; Barros-Souza, Y.; Maianne, M.; Borges, L. M. Radiations of the sensitive plants: exploring the influence of abiotic niche in diversification of the mega-genus *Mimosa* L. (Leguminosae). 1st Brazilian Congress of Evolutionary Biology, Curitiba (PR), Brazil

2023 **Barduzzi, R. F**.; Liau-Kang, S.; Barros-Souza, Y.; Borges, L. M. Evolution of pollen morphology in the Stryphnodendron clade (Fabaceae, Caesalpinioideae). 73th Brazilian Congress of Botany, Belém (PA), Brazil

2022 **Barduzzi, R. F**.; Liau-Kang, S. and Borges, L. M. Pollen disparity in the Piptadenia group (Leguminosae): differences between lineages and ecological groups. 2022. 72th Brazilian Congress of Botany, Brazil

TEACHING EXPERIENCE

Academia

2026 Pollen grains: Understanding their morphology and importance for taxonomy — 2nd edition, **Short course** at the XVIII Universidade Federal de São Carlos Biology Week (*Semana Acadêmica da Biologia*), Lead organizer

2025 Phytogeographic Analyses in R, **Short course** at the IV Rio–São Paulo Botany Meeting (*Jornada Rio–São Paulo de Botânica*), co-organizer.

2024 Pollen grains: Understanding their morphology and importance for taxonomy — 1st edition, **Workshop** at the XVI Universidade Federal de São Carlos Biology Week (*Semana Acadêmica da Biologia*), Lead organizer

2023 Behavioral Biology, **Undergraduate course**, Universidade Federal de São Carlos, Teaching Assistant

Student supervision

2026-present Diogo Mantovanelli Rabetti (Universidade Federal de São Carlos – **Undergraduate Research**). Co-supervision with Leonardo M. Borges

2025-2026 Felipe Ribeiro Alves Selotto (Universidade Federal de São Carlos - **Bachelor's Thesis**). Co-supervision with Leonardo M. Borges

PROFESSIONAL AFFILIATIONS

Society for the Study of Evolution (2024–)
Brazilian Society of Evolutionary Biology (2024–)

ACADEMIC SERVICE

Peer review
Evolutionary Ecology

Service to Universidade Federal de São Carlos
Student Representative, Undergraduate Program Council in Biological Sciences (2019-2023). Represented undergraduate students in curricular and academic matters and participated in program council meetings and discussions.

LANGUAGES

Portuguese (native); English (fluent)

FIELDWORK EXPERIENCE

Central Brazil (2023); Southwest Brazil (2024, 2026). Collected and prepared botanical specimens for herbarium deposition and collected data on plant–pollinator interactions.